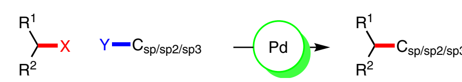
Synthesis Z. Qureshi et al. Short Review

# Secondary Alkyl Groups in Palladium-Catalyzed Cross-Coupling Reactions

**Zafar Qureshi**
**Christina Toker**
**Mark Lautens***

Department of Chemistry, Davenport Research Laboratories, University of Toronto, 80 St. George St., Toronto ON M5S 3H6, Canada
mlautens@chem.utoronto.ca




This review is dedicated to Professor Dieter Enders on the occasion of his 70th birthday.

Received: 28.03.2016
Accepted: 02.04.2016
Published online: 27.04.2016
DOI: 10.1055/s-0035-1561625; Art ID: ss-2016-z0218-sr

**Abstract** Important classes of cross-coupling reactions are those leading to the incorporation of secondary alkyl groups. While early work in cross-coupling focused mainly on primary alkyl groups, the past decade has seen significant advances in introducing secondary and tertiary groups. This review will focus on the advances made in recent years referencing landmark publications where necessary. Expansion of scope, ease of reaction setup, and mechanistic considerations will be stressed. Applications of methodologies towards complex scaffolds and natural products will be also be evaluated.



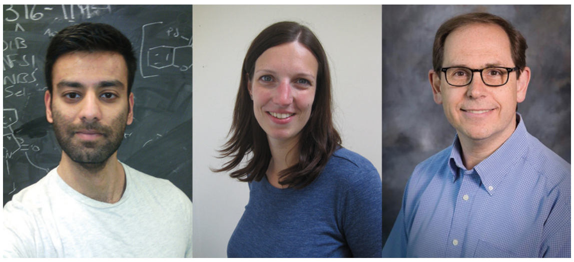
**Key words** palladium, homogenous catalysis, cross-coupling, secondary alkyl halides, C–H activation

## 1 Introduction

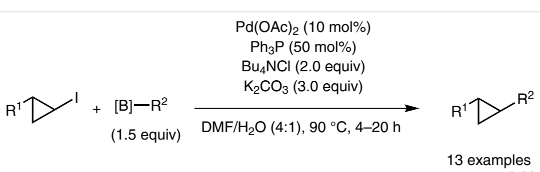
For aryl-aryl, aryl-alkyl or alkyl-alkyl couplings, the bulk of reports to date have utilized palladium. It is therefore no surprise that palladium has similarly been the focus in the area of secondary alkyl cross-couplings.[1] While aryl-aryl couplings have now become mainstream processes, aryl-alkyl coupling reactions have inherent challenges that are not present in aryl couplings.[2] Oxidative addition of alkyl halides onto palladium is a higher energy process compared to aryl halides. Secondary alkyl–metal species are also much less stable than their aryl–metal counterparts, leading to an increase in side reactions such as β-hydride elimination and isomerization of the secondary group.[3] Elegant studies by Falck, Molander, Fu, Sigman, and Catellani have bypassed some of these limitations based on catalyst and ligand design.[4] The most recent advances in palladium catalysis led to an immense number of reports with the coupling of secondary alkyl groups as the focus. Seminal reports focused on reactivity whereas more recent work has moved towards the enantioselective reactions of secondary alkyl groups in palladium catalysis. In this review we will focus on the latest advances referencing the landmark publications which initiated the corresponding fields.

## 2.1 Suzuki Coupling

Nucleophilic secondary alkyl groups have been reported as efficient coupling partners in palladium-catalyzed Suzuki reactions. To date the most prominent secondary alkylboron reagent used in palladium catalysis are cyclopropylboranes. These compounds bypass one of the most challenging barriers to secondary alkyl cross-couplings: β-hydride elimination. The earliest reports of the use of cyclopropylboronic acids in Suzuki reactions were independently published by Marsden,[5] Deng,[6] and Charette.[7] A recently published review and book chapter have taken an in-depth look at Suzuki reactions.[8] Moreover the group of Molander published a perspective on organotrifluoroborates in transition-metal-catalyzed cross-couplings with a strong focus on palladium.[9] The earliest example of the use of electrophilic secondary alkyl groups in Suzuki couplings was reported in 1996.[10] Charette and co-workers coupled iodocyclopropanes with vinylboronic esters with $Pd(OAc)_2/Ph_3P$ and $K_2CO_3$ as base in a 4:1 mixture of DMF/$H_2O$ (Scheme 1). High yields of differentially substituted vinylcyclopropanes could be obtained with the addition of $Bu_4NCl$.

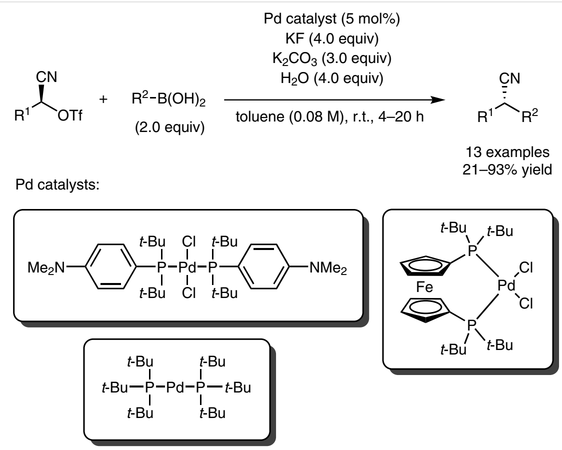


**Zafar Qureshi** was born in Karachi, Pakistan. He obtained his B.Sc. from the University of British Columbia in 2011. Under the supervision of Prof. Marco Ciufolini he worked on the synthesis of bioactive compounds and their analogues. Currently he is in his last year at the University of Toronto, working on obtaining his Ph.D. under the supervision of Prof. Mark Lautens. His research interests include transition-metal-catalyzed C–H functionalization, multicomponent-multicatalytic reactions and the synthesis of biologically active compounds.
**Christina Toker** was born in Cuxhaven, Germany, in 1983. She received her Diploma in chemistry in 2010 from the University of Bremen, Germany. In the same year she started her doctorate studies in the group of Prof. Markus Kalesse at the Helmholtz Centre for Infection Research, which she had already joined for her external diploma thesis, and received her Ph.D. from the Leibniz Universität Hannover, Germany, in 2014. Under the supervision of Prof. Mark Lautens she started her postdoctoral fellowship at the University of Toronto, Canada, in 2014. Her research interests are the total synthesis of biologically active natural products and the development of new synthetic methods based on metal catalysis on the way to biologically active compounds.
**Mark Lautens** was born in Hamilton, Ontario, Canada. He obtained his B.Sc. degree from the University of Guelph followed by doctoral studies at the University of Wisconsin-Madison under the direction of Barry M. Trost where he discovered Mo-catalyzed allylic alkylation and the Pd enyne cycloisomerization. In 1985 he moved to Harvard University where he conducted his NSERC PDF with David A. Evans on studies directed toward the synthesis of bryostatin. He joined the University of Toronto in 1987 as an NSERC University Research Fellow and Assistant Professor. Since 1998 he has held an Endowed Chair, the AstraZeneca Professor of Organic Synthesis, and from 2003–2013 he was named an NSERC/Merck Frosst Industrial Research Chair in New Medicinal Agents via Catalytic Reactions. In 2012, he was appointed as University Professor, the highest rank at the University of Toronto and in 2013 was selected as the J. Bryan Jones Distinguished Professor. Most recently he was awarded Officer of the Order of Canada, the highest civilian honor in Canada. His current research interests are in multicomponent-multicatalytic reactions, isomerization reactions, and applications of catalysis in the synthesis of bioactive compounds.



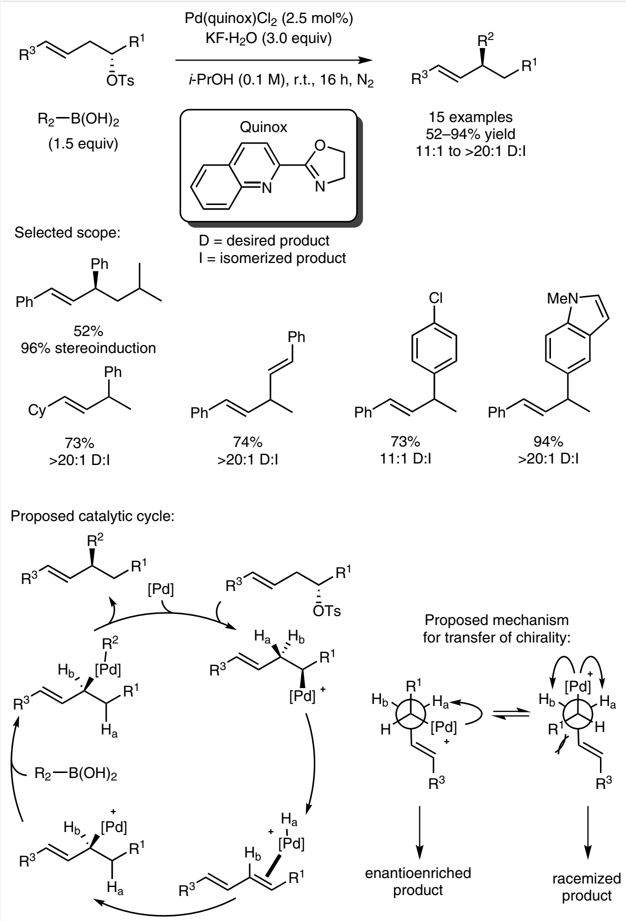

**Scheme 1** Cyclopropyl iodide coupling

In a study of less biased systems, Falck in 2010 utilized α-cyanohydrin alkyl triflates in the palladium-catalyzed Suzuki reaction (Scheme 2).[11] Although the electron-withdrawing nature of the cyano group would decrease the barrier to oxidative addition of the alkyl triflate, the linear substrate can undergo β-hydride elimination. Indeed, the only phosphine which promoted this coupling without elimination was the bulky electron-rich phosphine, *t*-$Bu_3P$. The optimized conditions coupled aryl- and vinylboronic acids using pre-formed palladium/ligand complexes and KF as base in toluene at room temperature. Using chiral cyano triflates led to products with inversion of stereochemistry. Since reductive elimination has been shown to go with retention of configuration[12] the oxidative addition of the palladium(0) catalyst must proceed with inversion of the alkyl triflate, pointing towards an $S_N2$ type mechanism.

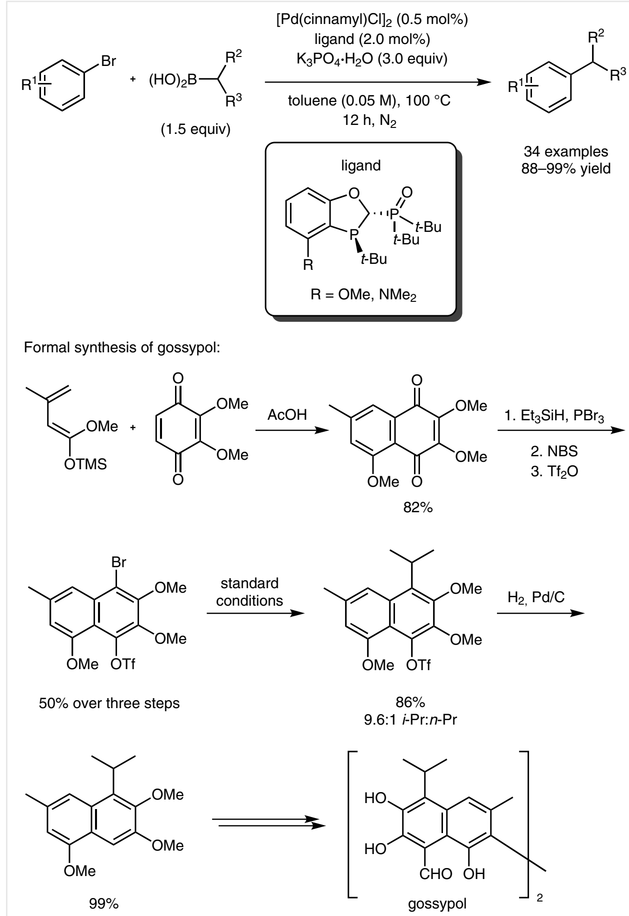



**Scheme 2** Linear alkyl triflate coupling

In 2012, Sigman *et al.* reported the synthesis of branched alkyl products using homoallylic alkyl tosylates and arylboronic acids in the presence of a Pd–quinox system (Scheme 3).[13] The reaction could be conducted at room temperature overnight. The scope in terms of boronic acid was broad, giving good to excellent yields with *ortho*- and *para*-substituted arylboron reagents. Aniline- and indole-substituted boronic acids also gave excellent yields. With alkenylboronic acids, skipped dienes could be generated. In all cases the ratio of the desired to isomerized product was greater than 10:1, with many giving products with >20:1 selectivity. Variations on the homoallylic tosylates were also tolerated. The use of chiral secondary homoallylic tosylates provided the opposite stereochemistry of the product, in line with Falck's observations of an inversion of stereochemistry during the oxidative addition step. The authors rationalize these findings in their proposed mechanism which involves coordination of the alkene, oxidative addition of the tosylate via $S_N2$, β-hydride elimination, re-insertion followed by cross-coupling. In 2014, a follow up report showed that a switch from isopropyl alcohol as solvent and KF·$H_2O$ as base to a mixture of THF/$H_2O$ and $Cs_2CO_3$ led to improved yields.[14]



Scheme 3 Linear alkyl tosylate coupling



Scheme 4 Secondary alkylboronic acid coupling

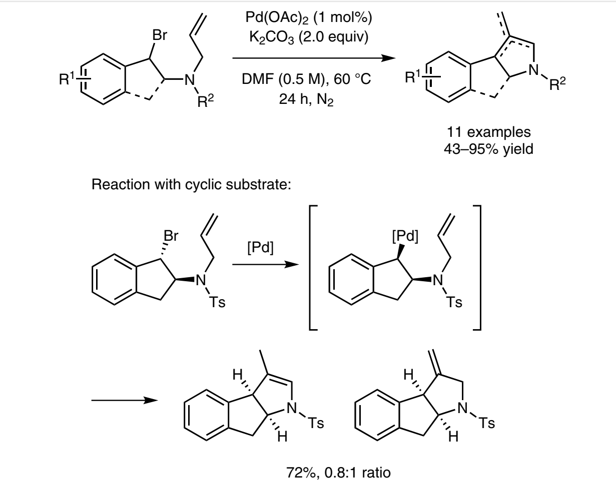
Generating 1,2,3-trisubstituted aromatic compounds has been a significant challenge due to the steric repulsion between the substituents on the aryl ring. However, due to the vast number of natural products containing such moieties it is imperative to develop methods to access these motifs. Tang and co-workers provided a method for the introduction of isopropyl- and isobutylboronic acids on *ortho*-substituted aromatic halides generating sterically hindered aryl compounds (Scheme 4).[15] The authors were able to showcase the utility of their methodology by a formal synthesis of gossypol, a male antifertility agent and PAF antagonist.[16]

## 2.2 Mizoroki–Heck Coupling

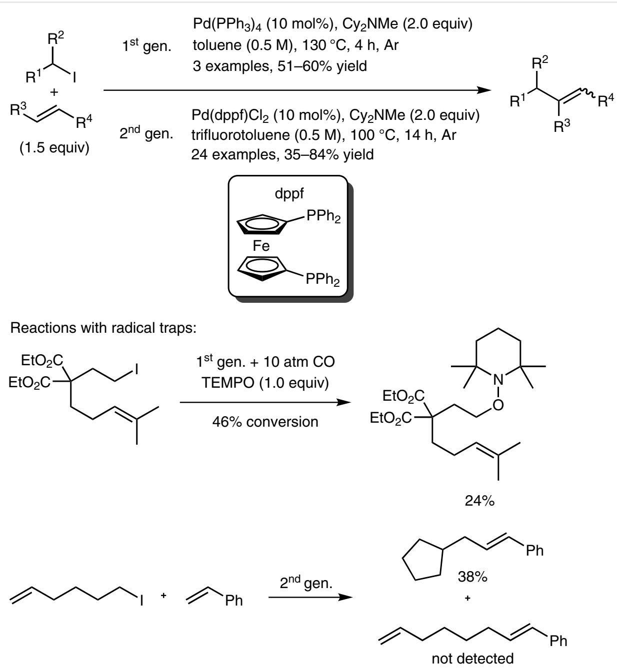
Since the first reports of the coupling of alkenes and aryl halides, the Mizoroki–Heck reaction has been one of the cornerstones of transition-metal-catalyzed reactions.[17] Since its discovery many developments have been reported including asymmetric variants.[18] However, only recently have secondary alkyl halides been utilized as electrophiles.

In 2011, Pan and co-workers[19] took a ligand-free approach for the synthesis of pyrroline derivatives in an intramolecular reaction (Scheme 5). $Pd(OAc)_2$ as catalyst and $K_2CO_3$ as base in DMF at 60 °C for 24 h delivered substituted pyrrolines in good yields, however in some cases double bond migration was observed. Reaction with a stereodefined substrate reinforced an $S_N2$ type mechanism.

At the same time Alexanian *et al.* reported the use of cyclohexyl iodide in an intermolecular Heck reaction (Scheme 6).[20a] The reaction was shown to tolerate both ketone and alcohol functional groups in the presence of $Pd(PPh_3)_4$ in toluene at 130 °C. The follow up in 2014 used a Pd–dppf system to expand the scope to include several cyclic and acyclic substrates coupling with electron deficient mono- and di-substituted alkenes.[20b] Generally a mixture of *E*/*Z* isomers was obtained. With the use of radical traps (TEMPO and tethered alkene), the authors concluded the transformation progressed via a radical rather than an ionic pathway; a stark difference from the $S_N2$-type mechanisms proposed for previous reactions.




**Scheme 5** Secondary alkyl bromide coupling

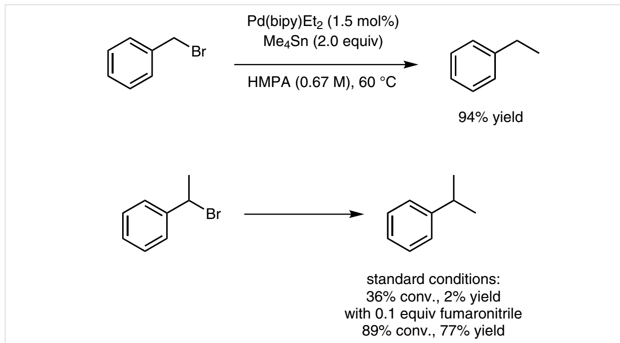



**Scheme 6** Secondary alkyl iodide coupling

Zhou and co-workers also identified cyclohexyl iodide as a competent coupling partner with a Pd–dppf combination.[21] Similarly, the trapping experiments with TEMPO and exogenous alkenes points towards a radical rather than an ionic mechanism.

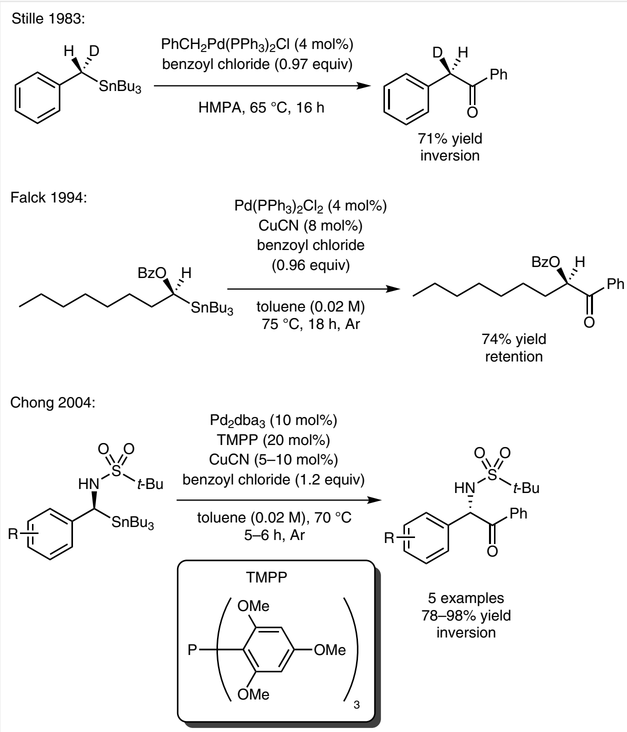
## 2.3 Stille Coupling

Although organotin compounds are known carcinogens[22] they have enjoyed a rich history in palladium catalysis due to their reactivity as nucleophiles.[23] The first example of secondary alkyl halides, used in a Stille coupling, was reported by Sustmann and co-workers in 1986.[24] Coupling of benzyl bromide with tetramethyltin proceeded smoothly with 1.5 mol% Pd(bipy)Et$_2$ in HMPA at 60 °C overnight to give ethylbenzene in 94% yield (Scheme 7). 1-Phenylethyl bromide, however failed to produce the cross-coupling product with Pd(bipy)Et$_2$ alone, generating styrene as a by-product. The addition of 0.1 equivalents of fumaronitrile was necessary for maximum yield (77%) of the coupling product. The exogenous alkene is believed to occupy the empty coordination site on palladium required for β-hydride elimination.



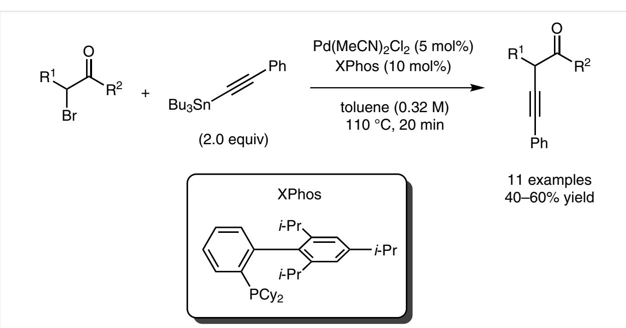

**Scheme 7** Effect of fumaronitrile on coupling 1-phenylethyl bromide

Since this discovery, Falck and co-workers expanded the scope to different classes of stannanes. Their first report included vinyl- and arylstannanes coupling to α-halo (thio)ethers.[25] A Pd/Cu co-catalyst system was introduced to promote the coupling of α-alkoxy- and α-aminostannanes[26] and stannyl glucopyranosides[27] to acyl chlorides. To examine the stereochemical aspects of the coupling of α-alkoxystannanes, (*S*)-[1-(benzoyloxy)octyl]tributylstannane was prepared and coupled to benzoyl chloride (Scheme 8). Analysis of the product revealed retention of configuration, which is in contrast to the coupling of tetraalkylstannanes with benzoyl chlorides, which Stille reported to proceed with inversion of stereochemistry (Scheme 8).[28] Stereoretentive coupling of α-alkoxystannes was expanded to include aryl and vinyl bromides and iodides.[29] Using Pd/Cu combinations, the group of Chong studied the coupling of α-aminostannanes with benzoyl chloride.[30] Unlike the α-alkoxystannanes, the α-aminostannanes reacted in a similar fashion to tetraalkylstannanes, giving products with inversion of stereochemistry (Scheme 8). These observations suggest that when a benzylic stannane is utilized, inversion of stereochemistry results, perhaps pointing towards a switch in mechanisms between benzylic and alkylstannanes.



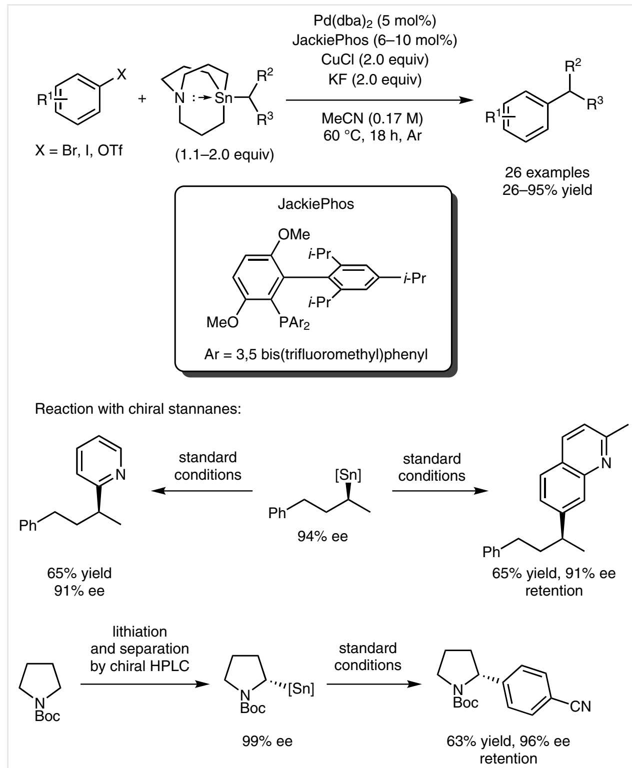

**Scheme 8** Benzylic versus non-benzylic alkylstannane coupling

In 2011, Connell and Kang showed alkynylstannanes were competent in the Stille coupling of secondary alkyl halides (Scheme 9).[31] α-Bromo esters could be coupled with tributyl(phenylethynyl)stannane in modest yields with a combination of Pd(MeCN)$_2$Cl$_2$ and the bulky monodentate phosphine XPhos.




**Scheme 9** Alkynylstannane coupling with α-bromo esters

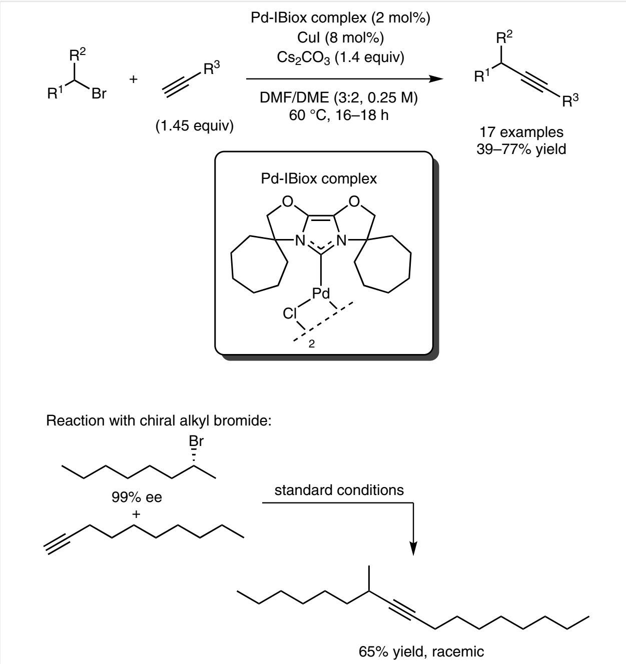
The most modern take on the coupling of chiral secondary alkylstannanes was demonstrated by Biscoe and co-workers (Scheme 10).[32] Utilizing azastannatranes developed by Vedejs and co-workers,[33] the authors were able to couple unactivated *s*-butylazastannatranes with halo(pseudohalo)-substituted arenes in the presence of Pd/JackiePhos as catalyst and CuCl as a mediator. No isomerization of the *sec*-butyl group to *n*-butyl was observed. The scope included electron-rich, electron-poor, and heteroaromatic electrophiles with yields ranging from 26–95%. Both cyclic and acyclic secondary alkyl azastannatranes containing ethers, amines, esters, and amides could be coupled efficiently under this protocol. Reaction with chiral alkylstannatranes[34] confirmed the retention of stereochemistry as observed by Falck for non-benzylic stannanes.



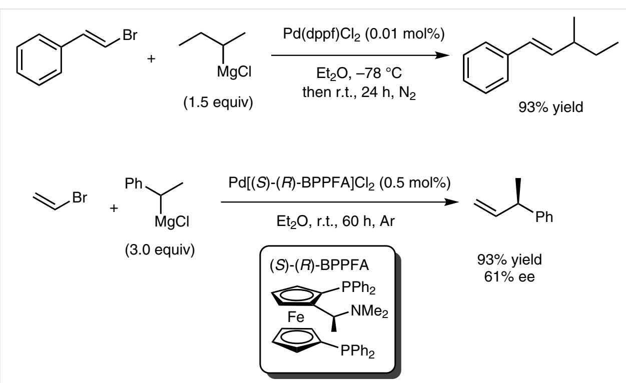

**Scheme 10** Azastannatrane coupling

## 2.4 Sonagashira Coupling

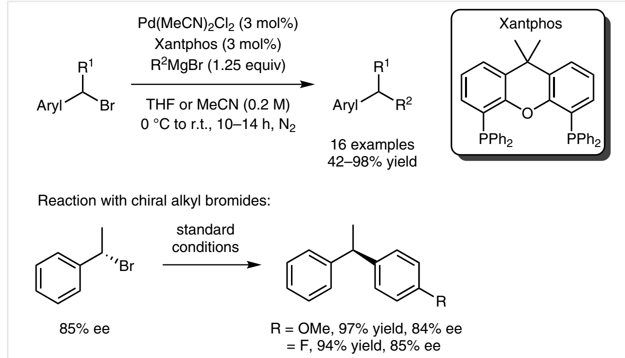
Glorius and co-workers reported the first example of a Sonagashira reaction with secondary alkyl bromides (Scheme 11).[35] It was found that NHC ligands[36] produced superior results when compared to phosphine ligands. The pre-formed Pd–IBiox complex with Cs$_2$CO$_3$ as base (3:2 mixture of DMF/DME at 60 °C) could couple cyclic and acyclic alkyl bromides with oct-1-yne efficiently. Yields ranged from 39–77%. Unfortunately the scope was limited in terms of the terminal alkyne. Neither phenylacetylene nor propargyl alcohol could be coupled. When stereochemically pure (*R*)-2-bromooctane was coupled to dec-1-yne the product was obtained in 65% yield as a racemate.




**Scheme 11** Secondary alkyl bromide coupling with terminal alkynes



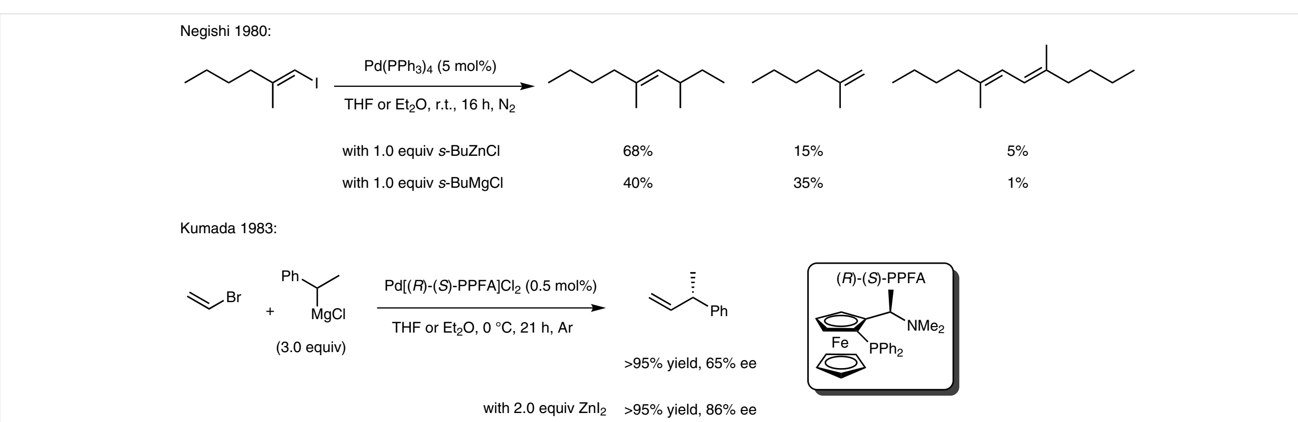

**Scheme 12** Asymmetric coupling of secondary alkyl Grignard reagents




**Scheme 13** Stereoinvertive coupling of 1-phenylethyl bromide with Grignard reagents

## 2.5 Kumada Coupling

The coupling of secondary alkyl Grignard reagents with aryl and vinyl bromides in the presence of palladium was accomplished by Kumada and co-workers in 1979.[37] Since then there have been numerous reports on the use of Grignard reagents in palladium-catalyzed cross-couplings.[38] In the coupling of β-bromostyrene and *s*-butylmagnesium chloride, $Pd(dppf)Cl_2$ loading as low as 0.01 mol% produced the coupled product in 93% yield after 24 h at room temperature (Scheme 12). Non-ferrocenyl based phosphine ligands (dppe, dppp, dppb, and $Ph_3P$) were completely ineffective, leading to incomplete conversion, isomerization, or dehalogenation. Having identified the ferrocenyl ligand class, the authors reported the first chiral coupling of secondary alkyl Grignard reagents with aryl and vinyl bromides.[39] Capitalizing on the stereochemical lability of organomagnesium reagents,[40] the authors were able to devise a dynamic kinetic resolution[41] of racemic substrates. The highest enantiomeric excess was achieved with the palladium complex of the (*S*)-(*R*)-BPPFA ligand. The coupling of 1-phenylethylmagnesium chloride with vinyl bromide was achieved with 0.5 mol% of the Pd/ligand complex in 93% yield and 61% ee (Scheme 12). It was discovered that the




**Scheme 14** Effect of zinc salts on stereoselectivity in secondary alkyl Grignard coupling

planar chirality of the ligand played the dominant role in obtaining higher enantiomeric excess rather than the point chirality. The nitrogen atom is also essential for high enantiomeric excess and is postulated to coordinate to magnesium during the transmetalation step.

In 2009, Carretero and co-workers reported the first coupling of secondary alkyl bromides with aryl and vinyl Grignard reagents (Scheme 13).[42] $Pd(MeCN)_2Cl_2$ with Xantphos worked best coupling 1-bromoethylbenzene with vinylmagnesium bromide in 98% yield after 7 hours at room temperature. With chiral 1-phenylethyl bromide inversion of stereochemistry was observed.

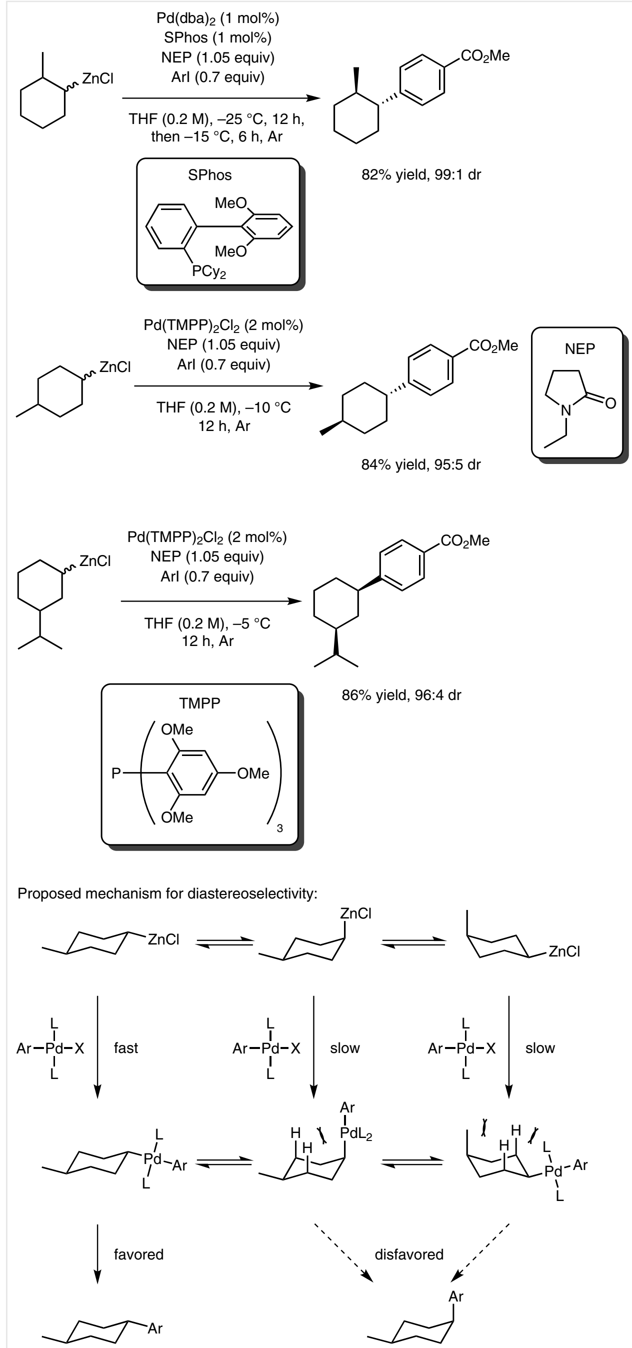
## 2.6 Negishi Coupling

It quickly became apparent that organozinc reagents were superior to Grignard reagents in palladium cross-couplings. Negishi's initial discovery in 1980 showed that coupling *s*-BuZnCl with a vinyl iodide provided a higher yield and lower byproduct formation when compared to *s*-BuMgCl (Scheme 14).[43] Kumada and co-workers published the enantioselective cross-coupling of 1-phenylethyl organometallic halides with vinyl bromide in the presence of their Pd[(*R*)-(*S*)-PPFA]$Cl_2$ catalyst.[44] Whereas the magnesium counterpart provided the coupling product in 65% ee, the use of zinc salts gave the product in 86% ee.

In 2010, Knochel and co-workers devised a clever diastereoselective coupling of substituted cyclohexylzinc reagents (Scheme 15).[45] 2-, 3-, and 4-Substituted cyclohexylzinc chlorides were coupled to aryl iodides. Remarkably high diastereomeric ratios were obtained for all substrates tested. The 2- and 4-substitution patterns furnished the *trans*-coupled products whereas the 3-substituted substrates gave the *cis* products. Using DFT and $^1H^{31}P$ HMBC NMR, the authors concluded that the zinc species interconverts between the axial and equatorial positions of the cyclohexyl ring, however the palladium species only resides at the equatorial position due to the repulsive forces between the ligand and the cyclohexyl ring.

Lipshutz and co-workers took a different approach to coupling organozinc reagents and aryl iodides with palladium. Using surfactants, they were able to couple alkyl bromides with aryl and vinyl bromides in water (Scheme 16).[46] The zinc dust selectively reacts with the alkyl bromide whereas the palladium catalyst reacts selectively with the aryl bromide. A diamine ligand was crucial to the success of this reaction as it aided the insertion of zinc into the alkyl bromide bond and also stabilized the newly formed organozinc species. Both cyclic and acyclic secondary alkyl bromides could be coupled in good yields without any isomerization.

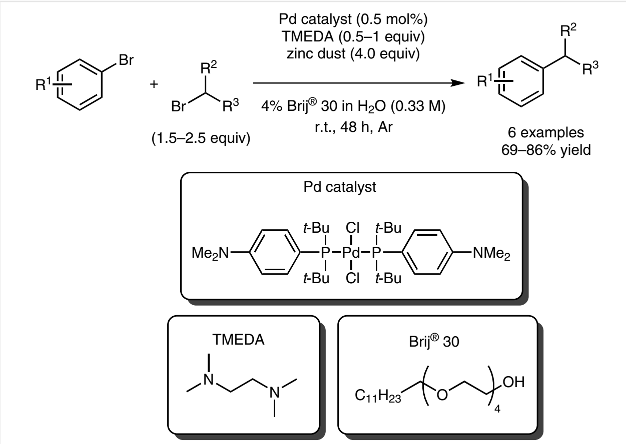
Through ligand design, the groups of Organ[47] and Buchwald[48] expanded the scope of the Negishi reaction to include aryl and heteroaryl halides that were difficult to



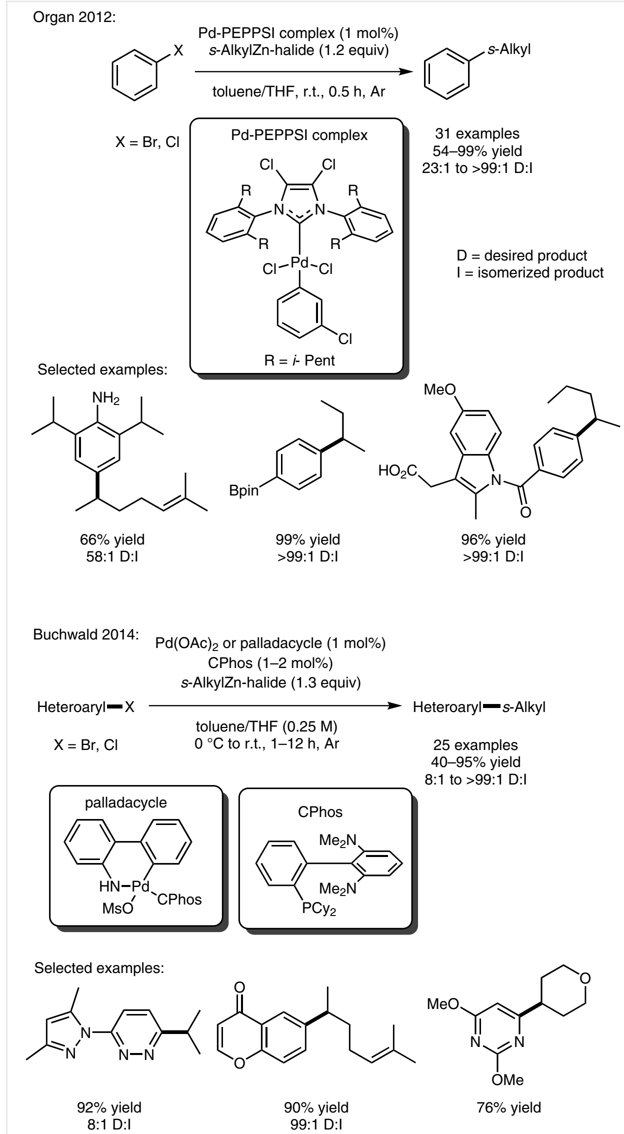

**Scheme 15** Stereoselective secondary alkylzinc coupling

couple using previously developed technologies (Scheme 17). Noteworthy substrates included unprotected acids, aniline, and boronic ester groups giving the coupled products in high yields.

The coupling of secondary alkylzinc reagents with acyl electrophiles has been reported as an efficient method for the preparation of unsymmetrical ketones. The groups of Harada/Oku,[49] Zhang,[50] and Seki[51] showed that acyl chlo-




**Scheme 16** Aryl bromides coupling to in situ generated secondary alkylzinc substrates




**Scheme 17** Secondary alkylzinc substrates coupling with preformed palladium complexes

rides, acyl anhydrides, and thioesters, respectively, are competent electrophiles in palladium couplings with secondary alkylzinc reagents (Scheme 18). Using a chiral monodentate phosphoramidite ligand, Reisman and co-workers showed the potential for an asymmetric coupling of an alkylzinc reagent with a thioester (Scheme 18).[52]

## 2.7 Hiyama Coupling

Hiyama and co-workers were able to couple aryl bromides with specially designed alkylsilicon substrates that contained a tertiary hydroxyl group.[53] Excellent yields were obtained for the coupling of isopropyl-, cyclopentyl-, and cyclohexylsilanes with both aryl and heteroaryl bromides (Scheme 19). The silanol could be recovered in good yield.

## 2.8 Lithium Coupling

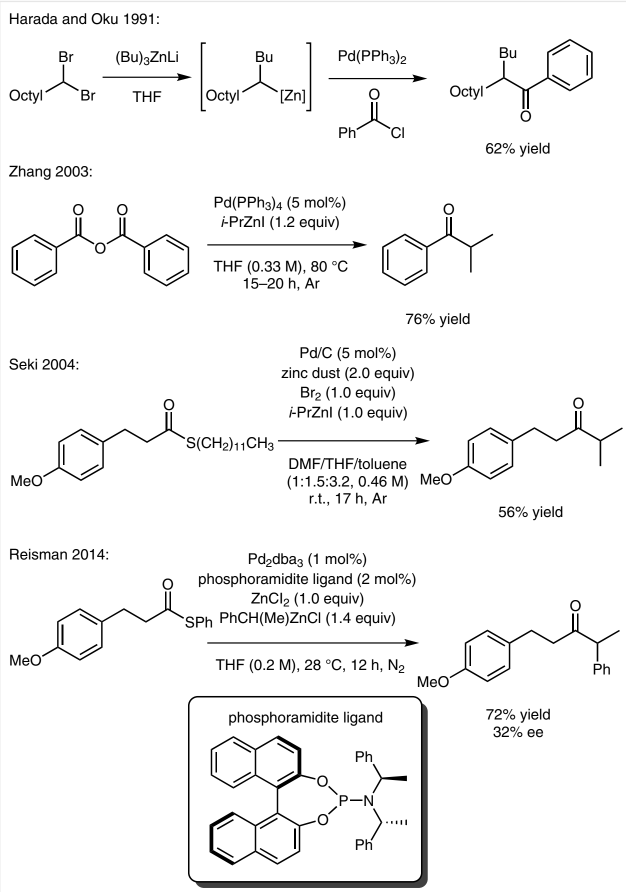
Organolithium reagents are highly reactive and thus are inherently unsuitable in many cross-couplings. However, Feringa and co-workers were able to couple both *i*-PrLi and *s*-BuLi with 4-$MeOC_6H_4Br$ with Pd(P*t*-$Bu_3$)$_2$ in toluene at room temperature in 78% and 89% yields, respectively (Scheme 20).[54] Later they expanded the scope to include a variety of secondary alkyllithium reagents coupling to both aryl and vinyl bromides.[55]

## 2.9 Titanium Coupling

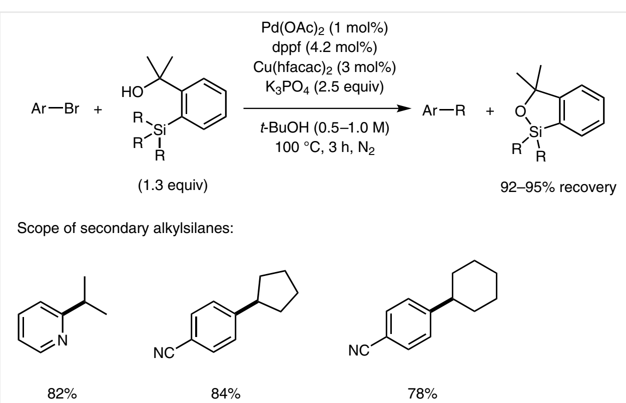
Organotitanium reagents have also been used in palladium cross-couplings with secondary alkyl groups. Gau and co-workers[56] reported one example of 1-phenylethyl bromide coupled with 4-$MeOC_6H_4Ti$(O*i*-Pr)$_3$in the presence of Pd$(OAc)_2$ and (*o*-Tol)$_3$P in toluene at room temperature in 2 hours (Scheme 21). Despite only partial conversion (79%), 70% of the product could be isolated.

## 2.10 C–H Activation

Pyridine-directed C–H activation has fast become an intensely studied field due to the selectivity imparted by the nitrogen atom.[57] Several scaffolds utilizing nitrogen as a directing group have been developed and utilized for the activation of secondary alkyl groups. The group of Daugulis first reported 8-aminoquinoline (AQ) as a suitable director in palladium catalysis.[58] Three examples using this scaffold were reported to undergo fast conversions with Pd$(OAc)_2$ loadings as low as 0.1 mol% with stoichiometric AgOAc as

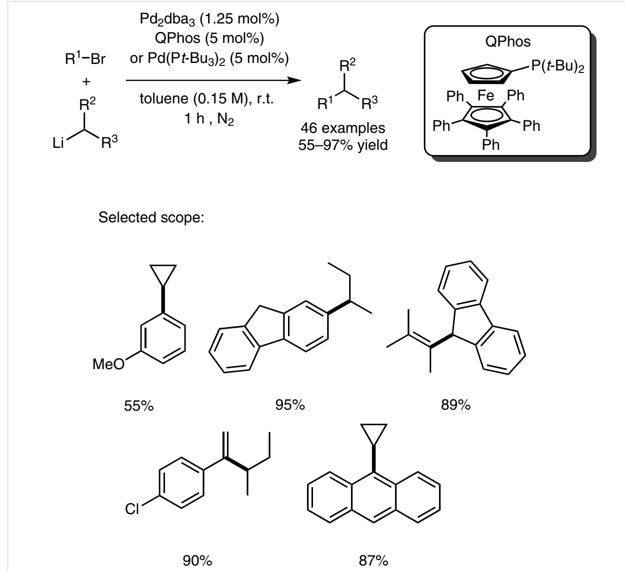



**Scheme 18** Secondary alkylzinc reagents coupling with acyl electrophiles

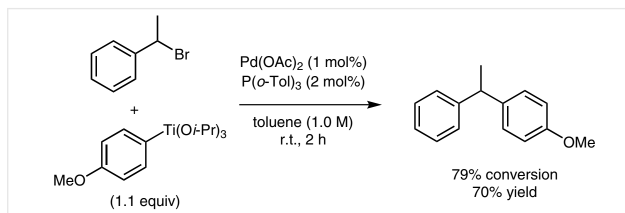



**Scheme 19** Aryl bromide coupling with secondary alkylsilanes

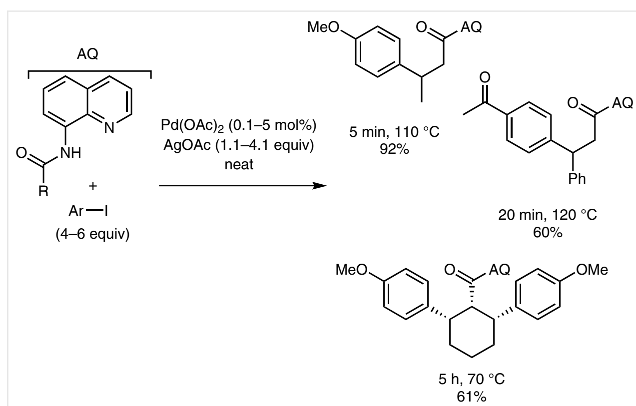
an additive (Scheme 22). The reaction was carried out neat at elevated temperatures with excess aryl iodide as the electrophile.




**Scheme 20** Secondary alkyllithium reagents coupling with aryl bromides



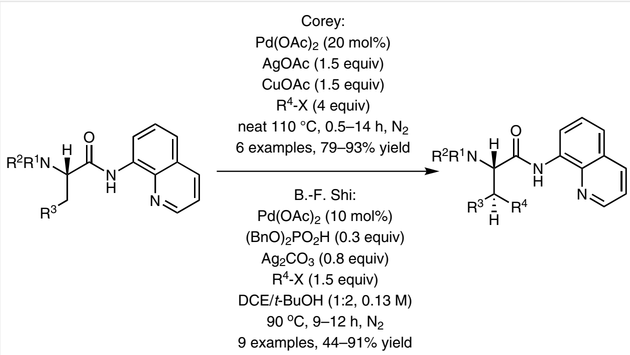

**Scheme 21** 1-Phenylethyl bromide coupling with aryltitanium reagent




**Scheme 22** Aminoquinoline directed C–H activation

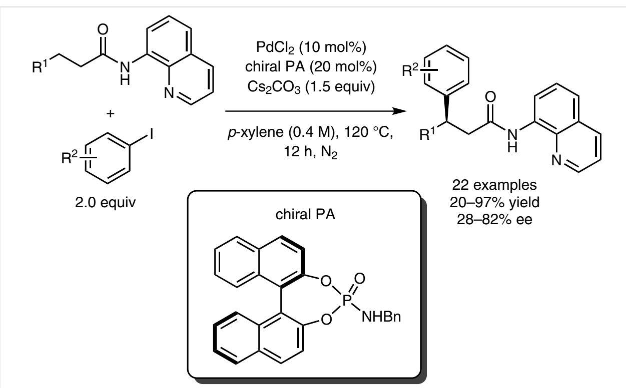
The groups of Corey[59] and B.-F. Shi[60] utilized this strategy to diastereoselectively functionalize alkyl groups of AQ adducts of amino acids (Scheme 23). A variety of unnatural chiral amino acids were synthesized in good yields.

Based on the nitrogen chelation to palladium, the groups of Chen[61] and Z.-J. Shi[62] expanded the field to include aryl and alkyl iodide electrophiles in both intra- and



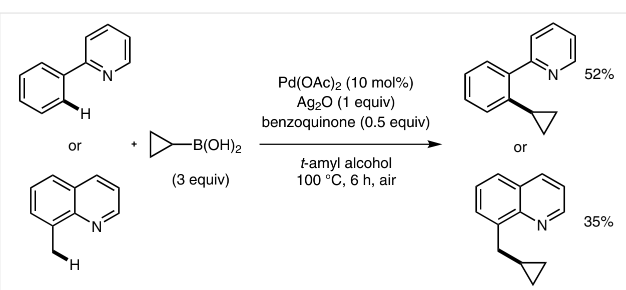

**Scheme 23** Synthesis of unnatural amino acids

intermolecular reactions. Using chiral phosphoric amides, Duan and co-workers were able to affect the enantioselective arylation based on the AQ directed C–H activation of secondary alkyl groups (Scheme 24).[63]



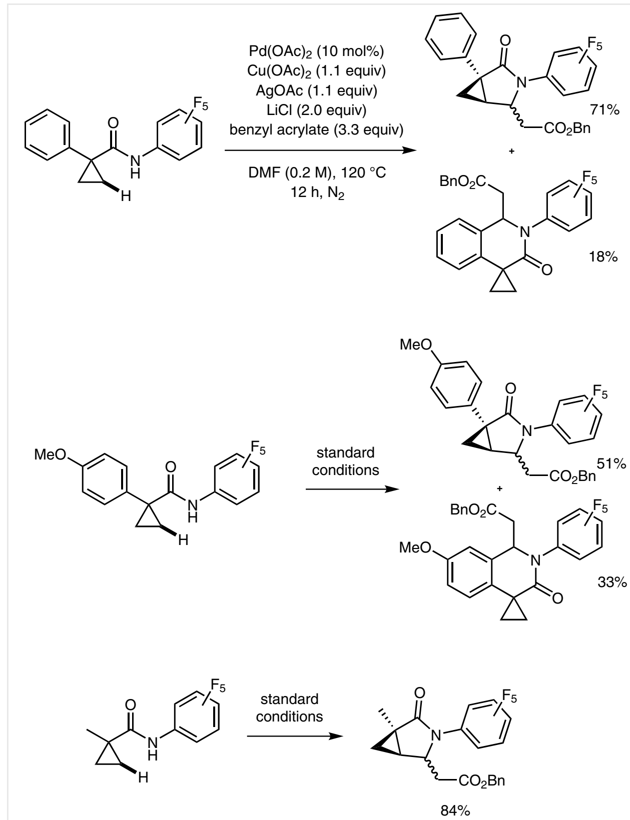

**Scheme 24** Stereoselective C–H arylation with chiral phosphoric amides

The first case of cyclopropyl groups used in palladium-catalyzed C–H functionalization reactions was directed by a pyridyl group.[64] In 2006, the group of Yu reported two examples of cyclopropylboronic acid coupling with 2-phenylpyridine and 8-methylquinoline giving the corresponding products in 52% and 35% yield, respectively (Scheme 25). Three equivalents of cyclopropylboronic acid were required along with stoichiometric amounts of $Ag_2O$.



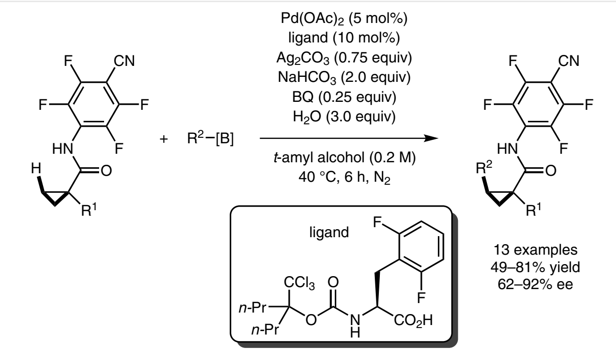

**Scheme 25** Cyclopropylboronic acid incorporation on nitrogenous heterocycles

With an electron-deficient aryl amide directing group three examples of the C–H activation of cyclopropanes were reported in 2010 (Scheme 26).[65] A combination of catalytic $Pd(OAc)_2$ and stoichiometric $Cu(OAc)_2$, AgOAc, and LiCl were required. Substrates with aryl groups were plagued with competing C–H activation of the aryl group. With electron-rich aromatic groups increased amounts of aryl C–H activation were observed. All substrates were obtained as mixtures of diastereomers.

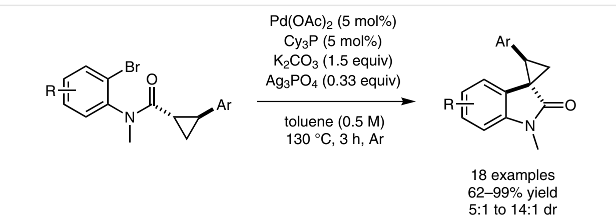



**Scheme 26** C–H activation of cyclopropanes using an amide directing group

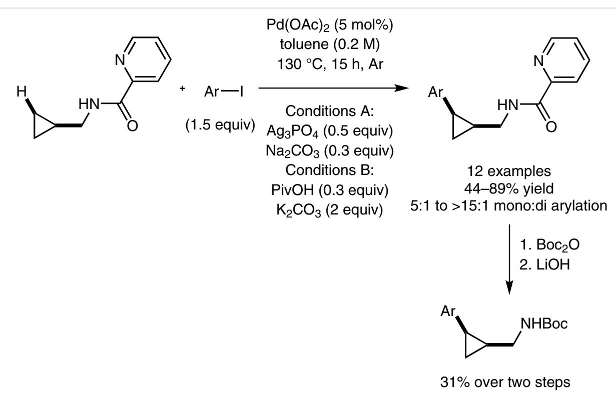
In 2011, a modified procedure was reported for the selective activation of cyclopropyl groups using an amide directing group (Scheme 27).[66] With $Pd(OAc)_2$ as the catalyst and a uniquely protected amino acid ligand, cyclopropanes could be functionalized with aryl-, vinyl-, and alkylboranes. However, only two examples where $R^1$ = aryl were reported, electron-deficient 4-chlorophenyl and 2,6-difluorophenyl, both of which would be biased against C–H activation on the aryl substituent. Yields of up to 81% and up to 92% ee could be attained without over functionalization. The authors found that the best procedure was to use a two-step protocol where additional catalyst, ligand, base, and reagents were added midway through the reaction.




**Scheme 27** Enantioselective C–H activation of cyclopropanes




**Scheme 28** Diastereoselective C–H activation of cyclopropanes

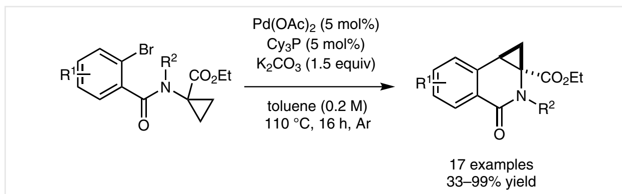



**Scheme 29** Intermolecular C–H arylation of cyclopropanes

Charette and co-workers expanded this methodology for the diastereoselective synthesis of spirocyclic oxindoles in 2013 (Scheme 28).[67] As with Yu's procedure, silver was a crucial additive.

Charette and co-workers also published an intermolecular variant, utilizing a picolinamide directing group (Scheme 29).[68] In this report both silver and pivalic acid could mediate the reaction. Several electron-rich, electron-poor, and heteroaryl iodides could be coupled. The reaction was shown to be efficient on gram scale and the picolinamide directing group could be removed in a two-step protocol.

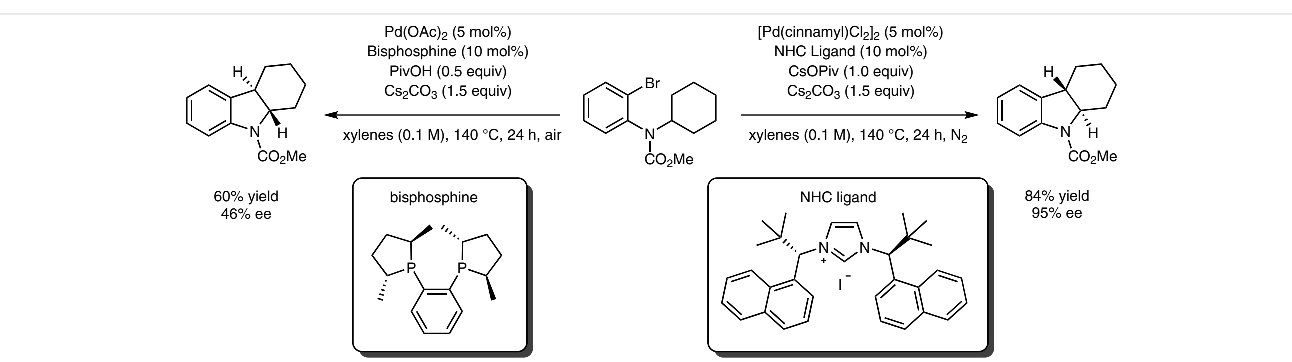
Their most recent protocol eliminates the need for any silver or pivalate additives (Scheme 30).[69] $Pd(OAc)_2$ with $Cy_3P$ as ligand and $K_2CO_3$ as base in toluene at 110 °C were sufficient to affect the arylation of cyclopropanes, even on a gram scale.



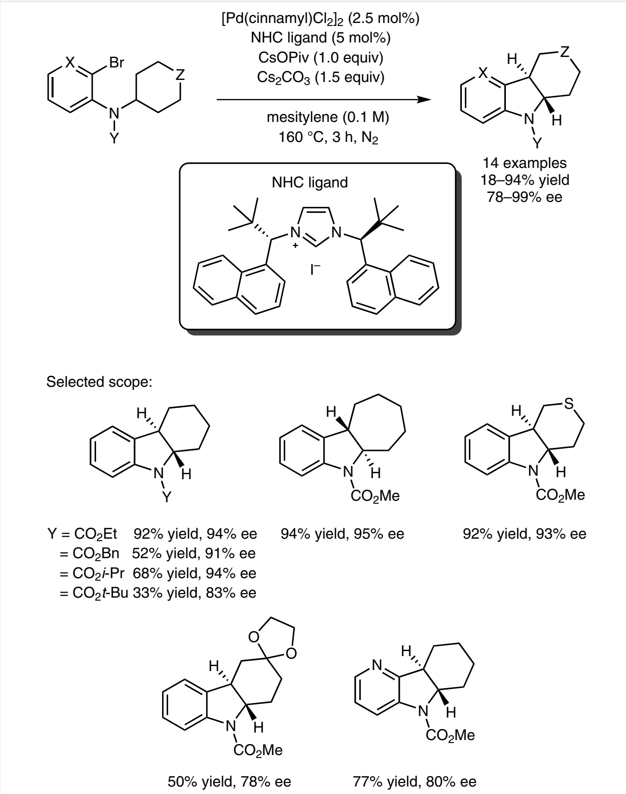

**Scheme 30** Silver-free C–H arylation of cyclopropanes

In 2011, the groups of Kündig and Kagan reported the enantioselective synthesis of fused indoline structures using bulky NHC and bisphosphine ligands, respectively (Scheme 31).[70] Good yields were obtained with both ligands, however the NHC ligand class was superior in terms of enantioselectivites.

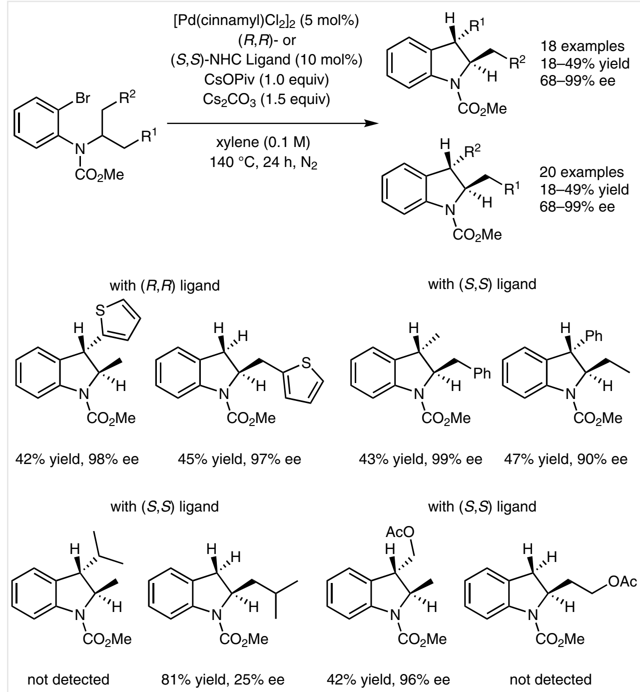
Subsequently, Kündig and co-workers expanded the scope to include other cyclic indolines with exquisitely high enantiomeric excess (Scheme 32).[71] Higher yields were observed with a smaller carbmate protecting group on nitrogen. The alkyl ring could be expanded to a 7-membered ring giving excellent yield and enantioselectivity. Tetrahydropyran, tetrahydrothiopyran, and piperidine derivatives



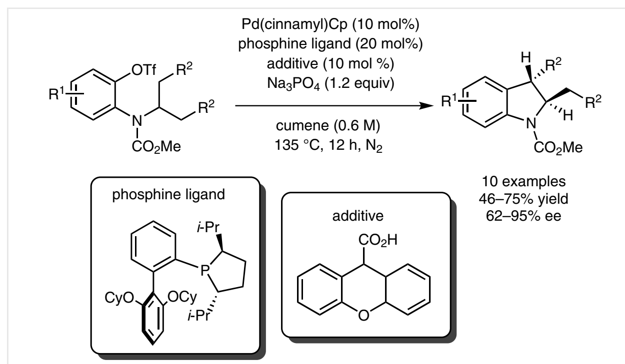

**Scheme 31** Enantioselective synthesis of indolines with bisphosphine and NHC ligands



**Scheme 32** Enantioselective synthesis of indolines using NHC ligands



**Scheme 33** Stereoselective kinetic resolution of racemic aryl bromides



**Scheme 34** Enantioselective synthesis of indolines from aryl triflates using biaryl phosphine ligands

were also competent. Both ketals and pyridine derivatives however, gave lower yields and enantioselectivity.

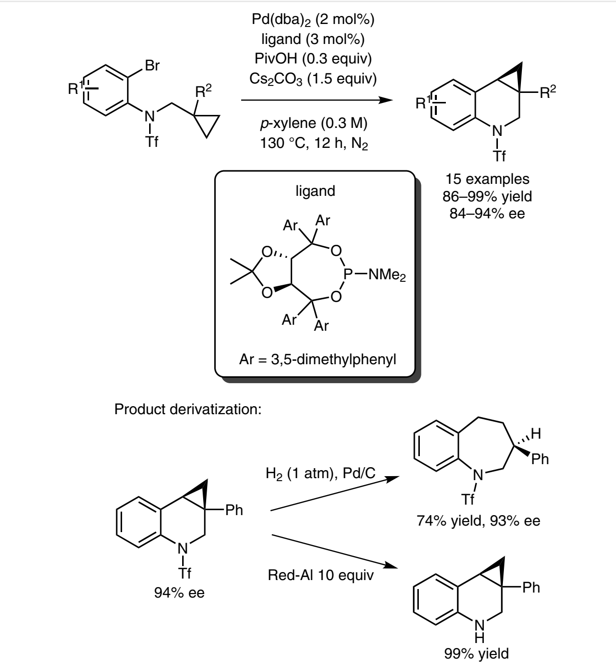
The Pd–NHC catalyst system was able to separate acyclic racemic substrates into two differentially substituted indolines with high enantiomeric excesses for both products (Scheme 33).[72]

At the same time Cramer and co-workers synthesized similar compounds starting from aryl triflates instead of aryl bromides (Scheme 34).[73] Instead of NHC-based ligands, the authors introduced a new class of chiral biaryl monophospholane ligands based on the Buchwald scaffold.[74] Like the NHC ligands of Kündig, Cramer's ligands also displayed exceptionally high enantioselectivites for the C–H activation of secondary alkyl groups.

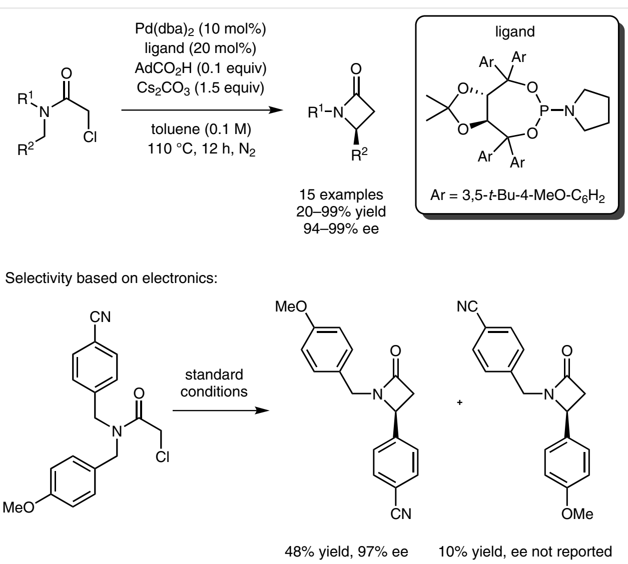
The same group also reported the enantioselective activation of tethered cyclopropyl groups in 2012 (Scheme 35).[75] In this case, the monophospholane ligands were completely ineffective and a bulky taddol phosphoramidite ligand was required to give conversion and selectivity.

The synthesis of smaller rings, such as β-lactams, were subsequently reported utilizing a similar taddol phosphoramidite ligand (Scheme 36).[76] The electronic properties of the alkyl groups on the nitrogen were important in determining regioselectivity. Electron-poor alkyl groups were activated faster than electron-rich groups, as evidenced by the nearly 5:1 ratio for the 4-cyanobenzyl and 4-methoxybenzyl groups.

In 2013, Fu and co-workers introduced a method for the coupling of pyridine *N*-oxides with secondary alkyl bromides (Scheme 37).[77] In the presence of 5 mol% $Pd(dppf)(OAc)_2$ and 2 equivalents of $Cs_2CO_3$ in toluene at 100 °C, a variety of secondary alkyl bromides could be installed at the 2-position of pyridine *N*-oxides including 4-, 5-, 6-, 7-, and 8-membered rings and heterocyclic alkyl bromides. Initial mechanistic studies point towards a radical process. A large-scale experiment followed by deprotection demonstrated the reaction was amenable to gram-scale synthesis of 2-alkylated pyridines.

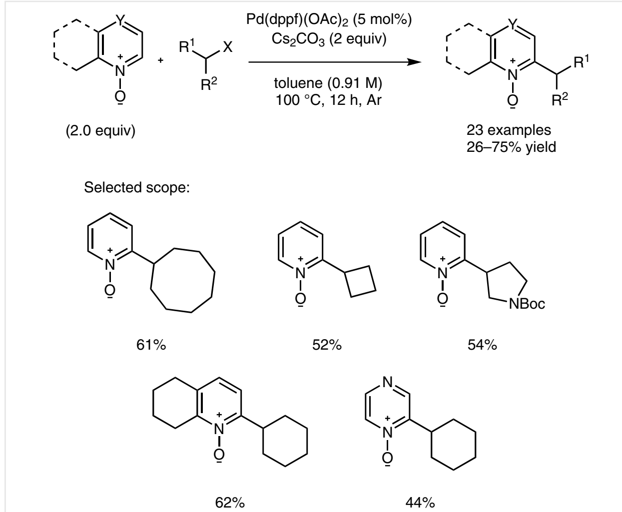



**Scheme 35** Enantioselective arylation of cyclopropanes using phosphoramidite ligands

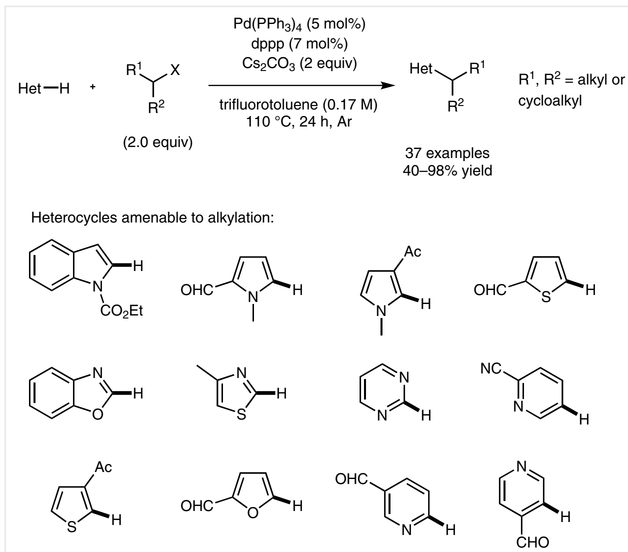



**Scheme 36** Enantioselective synthesis of β-lactams

Zhou and co-workers expanded this methodology to alkylate a variety of other heterocycles (Scheme 38).[78] Using $Pd(PPh_3)_4$ as catalyst and dppp as ligand with $Cs_2CO_3$ in trifluorotoluene at 110 °C for 24 hours, a similar array of alkyl iodides and bromides could be coupled. 5- and 6-membered azacycles, furans, thiophenes, and pyrroles and their benzo-fused derivatives were competent in the reaction giving yields between 40–98%.

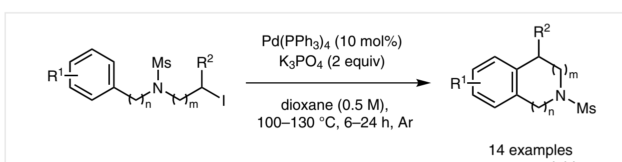



**Scheme 37** C–H activation of pyridine *N*-oxides



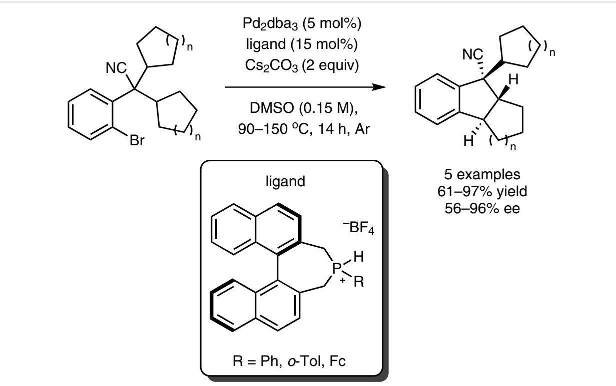

**Scheme 38** C–H activation of various heterocycles

The group of Alexanian reported on the synthesis of several nitrogenous heterocycles (Scheme 39).[79] With $Pd(PPh_3)_4$ as the catalyst and $K_3PO_4$ as the base in dioxane at elevated temperatures, substituted indolines, indoles, and pyrroles were synthesized in good yields without isomerization (β-hydride elimination/re-insertion) of the tethered secondary alkyl group.



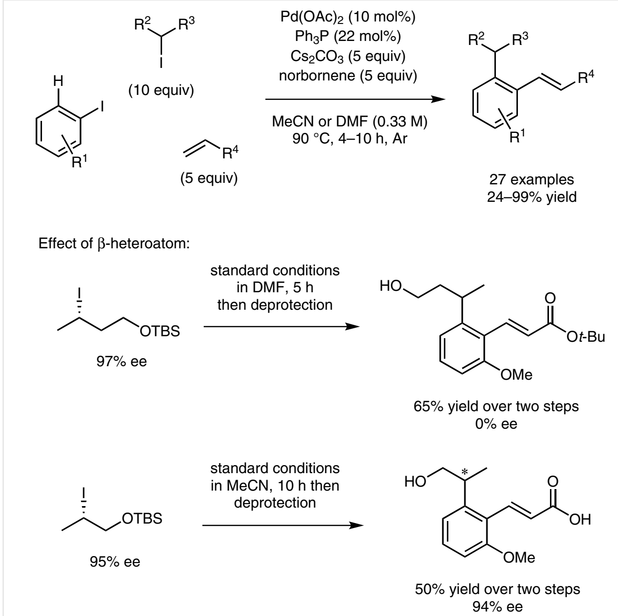

**Scheme 39** Synthesis of nitrogenous heterocycles

Baudoin and co-workers were able to develop new binepine ligands that allowed the diastereo- and enantioselective synthesis of highly congested indane scaffolds (Scheme 40).[80] Starting from aryl bromides with tethered alkyl groups, the palladium/binepine catalyst can choose one of two enantiotopic C–H bonds to activate. Various rings (5-, 6-, and 7-membered) along with pyrans and Boc-protected piperidines could be activated efficiently and with excellent enantioselectivites.




**Scheme 40** Stereoselective synthesis of indanes using binepine ligands

Since the initial report in 1997, the Catellani reaction[81] has gained popularity in the recent past.[82] But introducing secondary alkyl groups in the reaction has been a considerable challenge. Catellani first presented the capability to introduce isopropyl iodide in 1999.[83] The reaction was sluggish and did not go to completion. A synthetically more useful procedure was reported in 2000, however, similar to the previous case the reaction was slow, requiring 6 days to obtain the product in 71% yield.[84] The Lautens group utilized microwave irradiation to greatly accelerate the reaction. Synthetically useful yields could also be obtained for tethered alkyl halides.[85] The latest addition to this field was reported in 2015.[86] A variety of secondary alkyl iodides could be coupled with electronically flexible aryl iodides (Scheme 41). One limitation of the reaction was the requirement of a large excess (10 equiv) of alkylating agent. However, this excess could be recovered after the reaction. Chiral alkyl iodides were shown to readily racemize during the reaction, however the presence of a β-heteroatom was shown to stop this process. The reaction was also shown to be scalable, giving excellent yields on gram scale.




**Scheme 41** Incorporation of secondary alkyl halides in the Catellani reaction

## 3 Conclusion

The ability to introduce or link to secondary alkyl groups has been achieved using palladium catalysis allowing the generation of complex scaffolds. Structurally engineered substrates, ligand design, and control of reaction setup have allowed chemists to incorporate complex alkyl groups in palladium cross-couplings. No doubt the future will take these methodologies to the next step, using less expensive metals, reducing catalyst loadings, shortening reaction times, and increasing yield and selectivities.

## Acknowledgment

The authors thank the Natural Sciences and Engineering Research Council (NSERC), the University of Toronto, and Alphora Research Inc. for financial support. We thank the Canada Council for the Arts for a Killam Fellowship. Z. Q. thanks the Ontario Graduate Scholarship for funding. We also acknowledge Mark Levy for contributing in gathering references.

## References

(1) For recent reviews see: (a) Rudolph, A.; Lautens, M. *Angew. Chem. Int. Ed.* **2009**, *48*, 2656. (b) Kambe, N.; Iwasaki, T.; Terao, J. *Chem. Soc. Rev.* **2011**, *40*, 4937. (c) Jana, R.; Pathak, T. P.; Sigman, M. S. *Chem. Rev.* **2011**, *111*, 1417. (d) Swift, E. C.; Jarvo, E. R. *Tetrahedron* **2013**, *69*, 5799.
(2) (a) Pearson, R. G.; Figdore, P. E. *J. Am. Chem. Soc.* **1980**, *102*, 1541. (b) Ozawa, F.; Ito, T.; Yamamoto, A. *J. Am. Chem. Soc.* **1980**, *102*, 6457. (c) Luh, T.-Y.; Leung, M.-K.; Wong, K.-T. *Chem. Rev.* **2000**, *100*, 3187. (d) Frisch, A. C.; Beller, M. *Angew. Chem. Int. Ed.* **2005**, *44*, 674. (e) Hartwig, J. *Organotransition Metal Chemistry: From Bonding to Catalysis*; University Science Books: Sausalito, **2010**, 398–402.
(3) Stille, J. K.; Lau, K. S. Y. *Acc. Chem. Res.* **1977**, *10*, 434.
(4) Lundgren, R. J.; Stradiotto, M. *Chem. Eur. J.* **2012**, *18*, 9758.
(5) Hildebrand, J. P.; Marsden, S. P. *Synlett* **1996**, 893.

(6) Wang, X.-Z.; Deng, M.-Z. *J. Chem. Soc., Perkin Trans. 1* **1996**, 2663.
(7) Charette, A. B.; Perceira Da Freitas-Gil, R. *Tetrahedron Lett.* **1997**, *38*, 2809.
(8) (a) Leinori, D.; Aggarwal, V. K. *Angew. Chem. Int. Ed.* **2015**, *54*, 1082. (b) Crudden, C. M. *New Trends in Cross-Coupling: Theory and Applications*; Colacot, T. J., Ed.; RSC Catalysis Series No. 21: Cambridge, **2014**, 521–550.
(9) Molander, G. A. *J. Org. Chem.* **2015**, *80*, 7837.
(10) Charette, A. B.; Giroux, A. *J. Org. Chem.* **1996**, *61*, 8718.
(11) He, A.; Falck, J. R. *J. Am. Chem. Soc.* **2010**, *132*, 2524.
(12) Stille, J. K. *The Metal–Carbon Bond*; Vol. 2; Hartley, F. R.; Patai, S., Eds.; Wiley: New York, **1985**, 625–787.
(13) Stokes, B. J.; Opra, S. M.; Sigman, M. S. *J. Am. Chem. Soc.* **2012**, *134*, 11408.
(14) Stokes, B. J.; Bischoff, A. J.; Sigman, M. S. *Chem. Sci.* **2014**, *5*, 2336.
(15) Li, C.; Chen, T.; Li, B.; Xiao, G.; Tang, W. *Angew. Chem. Int. Ed.* **2015**, *54*, 3792.
(16) (a) Meyers, A. I.; Willemsen, J. J. *Chem. Commun.* **1997**, 1573. (b) Dodou, K.; Anderson, R. J.; Lough, W. J.; Small, D. A. P.; Shelley, M. D.; Groundwater, P. W. *Bioorg. Med. Chem.* **2005**, *13*, 4228. (c) Wei, J.; Jagt, D. L. V.; Royer, R. E.; Deck, L. M. *Tetrahedron Lett.* **2010**, *51*, 5757. (d) Li, L.; Liu, Y.; Wang, Q. *Eur. J. Org. Chem.* **2013**, 8014.
(17) (a) Heck, R. F. *J. Am. Chem. Soc.* **1968**, *90*, 5518. (b) Heck, R. F. *J. Am. Chem. Soc.* **1968**, *90*, 5526. (c) Mizoroki, T.; Mori, K.; Ozaki, A. *Bull. Chem. Soc. Jpn.* **1971**, *44*, 581. (d) Heck, R. F.; Nolley, J. P. *J. Org. Chem.* **1972**, *37*, 2320.
(18) McCartney, D.; Guiry, P. J. *Chem. Soc. Rev.* **2011**, *40*, 5122.
(19) Zhou, W.; An, G.; Zhang, G.; Han, J.; Pan, Y. *Org. Biomol. Chem.* **2011**, *9*, 5833.
(20) (a) Bloome, K. S.; McMahen, R. L.; Alexanian, E. J. *J. Am. Chem. Soc.* **2011**, *133*, 20146. (b) McMhon, C. M.; Alexanian, E. J. *Angew. Chem. Int. Ed.* **2014**, *53*, 5974.
(21) Zou, Y.; Zhou, J. *Chem. Commun.* **2014**, *50*, 3725.
(22) Kimbrough, R. D. *Environ. Health Perspect.* **1976**, *14*, 51.
(23) Cordovilla, C.; Bartolomé, C.; Martínez-Ilarduya, J. M.; Espinet, P. *ACS Catal.* **2015**, *5*, 3040.
(24) Sustmann, R.; Lau, J.; Zipp, M. *Tetrahedron Lett.* **1986**, *27*, 5207.
(25) Bhatt, R. K.; Shin, D.-S.; Falck, J. R. *Tetrahedron Lett.* **1992**, *33*, 4885.
(26) Ye, J.; Bhatt, R. K.; Falck, J. R. *J. Am. Chem. Soc.* **1994**, *116*, 1.
(27) Belosludtsev, Y. Y.; Bhatt, R. K.; Falck, J. R. *Tetrahedron Lett.* **1995**, *36*, 5881.
(28) Labadie, J. W.; Stille, J. K. *J. Am. Chem. Soc.* **1983**, *105*, 6129.
(29) Goli, M.; He, A.; Falck, J. R. *Org. Lett.* **2011**, *13*, 344.
(30) Kells, K. W.; Chong, J. M. *J. Am. Chem. Soc.* **2004**, *126*, 15666.
(31) Kang, J. Y.; Connell, B. T. *J. Org. Chem.* **2011**, *76*, 6856.
(32) Li, L.; Wang, C.-Y.; Huang, R.; Biscoe, M. R. *Nat. Chem.* **2013**, *5*, 607.
(33) (a) Vedejs, E.; Haight, A. R.; Moss, W. O. *J. Am. Chem. Soc.* **1992**, *114*, 6556. (b) Jensen, M. S.; Yang, C.; Hsiao, Y.; Rivera, N.; Wells, K. M.; Chung, J. Y. L.; Yasuda, N.; Hughes, D. L.; Reider, P. L. *Org. Lett.* **2000**, *2*, 1081.
(34) (a) Beak, P.; Kerrick, S. T.; Wu, S.; Chu, J. *J. Am. Chem. Soc.* **1994**, *116*, 3231. (b) Beak, P.; Basu, A.; Gallagher, D. J.; Park, Y. S.; Thayumanavan, S. *Acc. Chem. Res.* **1996**, *29*, 552.
(35) Altenhoff, G.; Würtz, S.; Glorius, F. *Tetrahedron Lett.* **2006**, *47*, 2925.
(36) Valente, C.; Çalimsiz, S.; Hoi, K. H.; Mallik, D.; Sayah, M.; Organ, M. G. *Angew. Chem. Int. Ed.* **2012**, *51*, 3314.
(37) Hayashi, T.; Konishi, M.; Kumada, M. *Tetrahedron Lett.* **1979**, *21*, 1871.
(38) (a) Hayashi, T.; Konishi, M.; Kobori, Y.; Kumada, M.; Higuchi, T.; Hirotsu, K. *J. Am. Chem. Soc.* **1984**, *106*, 158. (b) Yuan, K.; Scott, W. J. *Tetrahedron Lett.* **1989**, *30*, 4779. (c) Knappke, C. E. I.; Wangelin, A. J. *Chem. Soc. Rev.* **2011**, *40*, 4948.
(39) Hayashi, T.; Konishi, M.; Fukushima, M.; Mise, T.; Kagotani, M.; Tajika, M.; Kumada, M. *J. Am. Chem. Soc.* **1982**, *104*, 180.
(40) Hoffman, R. W. *Chem. Soc. Rev.* **2003**, *32*, 225.
(41) Huerta, F. F.; Minidis, A. B. E.; Bäckvall, J.-E. *Chem. Soc. Rev.* **2001**, *30*, 321.
(42) López-Pérez, A.; Adrio, J.; Carretero, J. C. *Org. Lett.* **2009**, *11*, 5514.
(43) Negishi, E.; Valente, L. F.; Kobayashi, M. *J. Am. Chem. Soc.* **1980**, *102*, 3298.
(44) Hayashi, T.; Hagihara, T.; Katsuro, Y.; Kumada, M. *Bull. Chem. Soc. Jpn.* **1983**, *56*, 363.
(45) (a) Thaler, T.; Haag, B.; Gavryushin, A.; Schober, K.; Hartmann, E.; Gschwind, R. M.; Zipse, H.; Mayer, P.; Knochel, P. *Nat. Chem.* **2010**, *2*, 125. (b) Moriya, K.; Knochel, P. *Org. Lett.* **2014**, *16*, 924.
(46) (a) Krasovskiy, A.; Duplais, C.; Lipshutz, B. H. *J. Am. Chem. Soc.* **2009**, *131*, 15592. (b) Krasovskiy, A.; Duplais, C.; Lipshutz, B. H. *Org. Lett.* **2010**, *12*, 4742. (c) Duplais, C.; Krasovskiy, A.; Lipshutz, B. H. *Organometallics* **2011**, *30*, 6090.
(47) (a) Çalimsiz, S.; Organ, M. G. *Chem. Commun.* **2011**, *47*, 5181. (b) Pompeo, M.; Froese, R. D. J.; Hadei, N.; Organ, M. G. *Angew. Chem. Int. Ed.* **2012**, *51*, 11354.
(48) (a) Han, C.; Buchwald, S. L. *J. Am. Chem. Soc.* **2009**, *131*, 7532. (b) Yang, Y.; Niedermann, K.; Han, C.; Buchwald, S. L. *Org. Lett.* **2014**, *16*, 4638.
(49) Harada, T.; Kotani, Y.; Katsuhira, T.; Oku, A. *Tetrahedron Lett.* **1991**, *32*, 1573.
(50) Wang, D.; Zhang, Z. *Org. Lett.* **2003**, *5*, 4545.
(51) Mori, Y.; Seki, M. *Tetrahedron Lett.* **2004**, *45*, 7343.
(52) Cherney, A. H.; Reisman, S. E. *Tetrahedron* **2014**, *70*, 3259.
(53) Nakao, Y.; Takeda, M.; Matsumoto, T.; Hiyama, T. *Angew. Chem. Int. Ed.* **2010**, *49*, 4447.
(54) Giannerini, M.; Fañanás-Mastral, M.; Feringa, B. L. *Nat. Chem.* **2013**, *5*, 667.
(55) Vila, C.; Giannerini, M.; Hornillos, V.; Fañanás-Mastral, M.; Feringa, B. L. *Chem. Sci.* **2014**, *5*, 1361.
(56) Chen, C.-R.; Zhou, S.; Biradar, D. B.; Gau, H.-M. *Adv. Synth. Catal.* **2010**, *352*, 1718.
(57) (a) Lyons, T. W.; Sanford, M. S. *Chem. Rev.* **2010**, *110*, 1147. (b) Giri, A.; Thapa, S.; Kafle, A. *Adv. Synth. Catal.* **2014**, *356*, 1395.
(58) Zaitsev, V. G.; Shabashov, D.; Daugulis, O. *J. Am. Chem. Soc.* **2005**, *127*, 13154.
(59) Reddy, B. V. S.; Reddy, L. R.; Corey, E. J. *Org. Lett.* **2006**, *8*, 3391.
(60) Chen, K.; Hu, F.; Zhang, S.-Q.; Shi, B.-F. *Chem. Sci.* **2013**, *4*, 3906.
(61) (a) Feng, Y.; Wang, Y.; Landgraf, B.; Chen, G. *Org. Lett.* **2010**, *12*, 3414. (b) Zhang, S.-Q.; He, G.; Nack, W. A.; Zhao, Y.; Li, Q.; Chen, G. *J. Am. Chem. Soc.* **2013**, *135*, 2124. (c) Zhang, S.-Q.; Li, Q.; He, G.; Nack, W. A.; Chen, G. *J. Am. Chem. Soc.* **2015**, *137*, 531.
(62) Chen, K.; Li, Z.-W.; Shen, P.-X.; Zhao, H.-W.; Shi, Z.-J. *Chem. Eur. J.* **2015**, *21*, 7389.
(63) Yan, S.-B.; Zhang, S.; Duan, W.-L. *Org. Lett.* **2015**, *17*, 2458.
(64) Chen, X.; Goodhue, C. E.; Yu, J.-Q. *J. Am. Chem. Soc.* **2006**, *128*, 12634.
(65) Wasa, M.; Engle, K. E.; Yu, J.-Q. *J. Am. Chem. Soc.* **2010**, *132*, 3680.
(66) Wasa, M.; Engle, K. E.; Lin, D. W.; Yoo, E. J.; Yu, J.-Q. *J. Am. Chem. Soc.* **2011**, *133*, 19598.
(67) Ladd, C. L.; Roman, D. S.; Charette, A. B. *Org. Lett.* **2013**, *15*, 1350.
(68) Roman, D. S.; Charette, A. B. *Org. Lett.* **2013**, *15*, 4394.
(69) Ladd, C. L.; Belouin, A. V.; Charette, A. B. *J. Org. Chem.* **2016**, *81*, 256.

(70) (a) Nakanishi, M.; Katayev, D.; Besnard, C.; Kündig, E. P. *Angew. Chem. Int. Ed.* **2011**, *50*, 7438. (b) Anas, S.; Cordi, A.; Kagan, H. B. *Chem. Commun.* **2011**, *47*, 11483.
(71) Larionov, E.; Nakanishi, M.; Katayev, D.; Besnard, C.; Kündig, E. P. *Chem. Sci.* **2013**, *4*, 1995.
(72) (a) Katayev, D.; Nakanishi, M.; Bürgi, T.; Kündig, E. P. *Chem. Sci.* **2012**, *3*, 1422. (b) Katayev, D.; Larionov, E.; Nakanishi, M.; Besnard, C.; Kündig, E. P. *Chem. Eur. J.* **2014**, *20*, 15021.
(73) Saget, T.; Lemouzy, S. J.; Cramer, N. *Angew. Chem. Int. Ed.* **2012**, *51*, 2238.
(74) (a) Surry, D. S.; Buchwald, S. L. *Angew. Chem. Int. Ed.* **2008**, *47*, 6338. (b) Martin, R.; Buchwald, S. L. *Acc. Chem. Res.* **2008**, *41*, 1461. (c) Surry, D. S.; Buchwald, S. L. *Chem. Sci.* **2011**, *2*, 27.
(75) Saget, T.; Cramer, N. *Angew. Chem. Int. Ed.* **2012**, *51*, 12842.
(76) Pedroni, J.; Boghi, M.; Saget, T.; Cramer, N. *Angew. Chem. Int. Ed.* **2014**, *53*, 9064.
(77) Xiao, B.; Liu, Z.-J.; Liu, L.; Fu, Y. *J. Am. Chem. Soc.* **2013**, *135*, 616.
(78) Wu, X.; See, J. W. T.; Xu, K.; Hirao, H.; Roger, J.; Hierso, J.-C.; Zhou, J. *Angew. Chem. Int. Ed.* **2014**, *53*, 13573.
(79) Venning, A. R. O.; Bohan, P. T.; Alexanian, E. J. *J. Am. Chem. Soc.* **2015**, *137*, 3731.
(80) Holstein, P. M.; Vogler, M.; Larini, P.; Pilet, G.; Clot, E.; Baudoin, O. *ACS. Catal.* **2015**, *5*, 4300.
(81) Catellani, M.; Frignani, F.; Rangoni, A. *Angew. Chem. Int. Ed.* **1997**, *36*, 119.
(82) (a) Lautens, M.; Alberico, D.; Bressy, C.; Fang, Y.-Q.; Mariampillai, B.; Wilhelm, T. *Pure Appl. Chem.* **2006**, *78*, 351. (b) Martins, A.; Mariampillai, B.; Lautens, M. *Top. Curr. Chem.* **2010**, *292*, 1. (c) Ye, J.; Lautens, M. *Nat. Chem.* **2015**, *7*, 863.
(83) Catellani, M.; Cugini, F. *Tetrahedron* **1999**, *55*, 6595.
(84) Catellani, M.; Motti, E.; Minari, M. *Chem. Commun.* **2000**, 157.
(85) (a) Rudolph, A.; Rackelmann, N.; Lautens, M. *Angew. Chem. Int. Ed.* **2007**, *46*, 1485. (b) Rudolph, A.; Rackelmann, N.; Turcotte-Savard, M.-O.; Lautens, M. *J. Org. Chem.* **2009**, *74*, 289.
(86) Qureshi, Z.; Schlundt, W.; Lautens, M. *Synthesis* **2015**, *47*, 2446.